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Delivery Document

AirNova

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Table of Contents

Context 3

Project Deliverables 3

Stakeholder Desires and Alignment 4

Next Steps and Future Considerations 5

Context

This document serves as the formal delivery of the AirNova: Sofia Air Quality Dashboard project. The primary objective was to develop a predictive data solution and a user-friendly interface to address the challenges faced by residents of Sofia, Bulgaria, due to unpredictable air quality, particularly concerning PM2.5 levels. The delivered solution is a fully functional web application that provides:

1. Real-time PM2.5 monitoring for Sofia.
2. Short-term hourly forecasts for PM2.5 levels.
3. A simple, color-coded health advisory system. This deliverable directly addresses the core needs identified during the stakeholder interview process, providing an actionable tool for planning daily activities.

Project Deliverables

The following components have been successfully developed and delivered as part of the AirNova project:

2.1. Predictive Data Model (AirNova Model)

A machine learning model was developed to forecast hourly PM2.5 concentrations in Sofia, Bulgaria.

Feature	Description
Model Type	Time-series regression model (likely Random Forest Regressor, based on methodology)
Data Sources	Historical Air Quality Data (PM2.5, PM10, CO, NO_2, SO_2, O_3) and Weather Data (Temperature, Humidity, Wind Speed/Direction, Pressure, Precipitation) for the last 2 years.
Prediction Horizon	Hourly forecasts for the short-term outlook.
Methodology	Data Collection, Cleaning & Preparation, Exploratory Data Analysis (EDA), Feature Engineering (Lag features, Cyclical encoding), and Model Evaluation.

2.2. User Interface: Sofia Air Quality Dashboard

A web-based dashboard was created to visualize the model's output and provide actionable information.

Dashboard Component	Description	Stakeholder Need Addressed
Current Air Quality	Real-time PM2.5 reading with a simple "Good" / "Moderate" / "Unhealthy" status.	Simple, reliable, and easy-to-understand data.
Forecasts (Local Time Based)	Clear display of the current PM2.5 forecast for the next few hours.	Knowing *beforehand* when pollution will be bad.
PM2.5 Trend Chart	A line chart showing the trend of PM2.5 (Last 24 Hours + Forecast).	Ability to click and see the data behind the alert.
Health Advisory	A prominent, color-coded banner (e.g., Green for Safe) based on current PM2.5 levels.	Simple dashboard with **color-coded alerts** (Green/Yellow/Red).
Detailed Outlook Table	A table listing hourly PM2.5 values, Air Quality status, and whether the value is Measured or Forecast.	Ability to click and see the data behind the alert.

Stakeholder Desires and Alignment

The project was guided by the specific needs and desires expressed by the stakeholder, Anna, during the interview process [1]. The delivered solution directly addresses these key requirements:

Stakeholder Desire [1]	AirNova Dashboard Feature	Alignment
"I wish I knew beforehand so I could prepare..." (Q1)	**Forecasts (Local Time Based)** and **PM2.5 Trend Chart**	Provides hourly predictions, allowing for proactive planning of outdoor activities.
"A simple dashboard with color-coded alerts, like green/yellow/red." (Q4)	**Health Advisory** and **Air Quality Status**	Uses a clear, color-coded system for immediate risk assessment.
"I' d like to click and see the data behind it." (Q4)	**PM2.5 Trend Chart** and **Detailed Outlook Table**	Offers granular, hourly data for users who want to investigate the forecast details.
"A personalized alert — maybe a notification that says 'PM2.5 will rise to unhealthy levels in your area in 2 hours.' " (Q5)	**Short-Term Outlook/Health Advisory**	The current dashboard provides the necessary data foundation for this feature. **Next Step:** Integration with a notification service (e.g., email, SMS, mobile app) is required to fully implement the *personalized alert* functionality.

Next Steps and Future Considerations

The current delivery provides a robust foundation for a public-facing air quality service. The following are recommended next steps:

1. Deployment: Deploy the dashboard from the local environment

2. Personalized Alert System: Develop and integrate a notification service (e.g., email, SMS, or mobile app push notification) to fully realize the stakeholder's desire for a personalized, proactive alert system.

3. Model Maintenance: Establish a routine for retraining the predictive model with new data to maintain accuracy and adapt to changing environmental conditions.